

Part I: Choose the best answer and write your choice on the space provided in capital letter! (9%)

A 1. If $A = (2m)i - (3m)j$ and $B = (1m)i - (2m)j$, then $A - 2B = ?$

~~A~~ (1m)j

B. (-1m)j

C. (4m)i - (7m)j

D. (4m)i + (1m)j

C 2. Four objects are situated on xy-plane as follows: a 1 kg object at the origin, 2kg object at +1cm on the x-axis, a 3kg object at +2cm on the y-axis, and a 4kg object at (+1cm, -2cm). Where is the center of mass of these objects?

A. (1,0)cm

B. (0.6, 1.4)cm

C. (0.6, -0.2)cm

D. (0.6, 0.2) cm

B 3. The "reaction" force does not cancel the "action" force because:

A. The action force is greater than the reaction force $\times$

B. They are on different bodies $\checkmark$

C. They are in the same direction $\times$

D. The reaction force exists only after the action force is removed

D 4. Which one of the following is true about fluid static?

A. All points at same level in a fluid have same pressure. $\checkmark$

B. Fluid pressure increases with increase in the depth of the fluid. $\checkmark$

C. Fluid pressure does not depend on the shape of the container. $\checkmark$

D. All

B 5. The small piston of a hydraulic lift has an area of 0.2 m². A car weighing 1.2×10^4 N sits on a rack mounted on the large piston. The large piston has an area of 0.9 m². How large force must be applied to the small piston to support the car?

A. 2.7×10^5 N

B. 2.7×10^3 N

C. 7×10^2 N

D. 3×10^4 N

B 6. A bead slides without friction around a loop (see fig). The bead is released from a height $h = 3.5R$. What is its speed at point A?

A. $\sqrt{3gR}$ m/s

B. $\sqrt{5gR}$ m/s

C. $\sqrt{7gR}$ m/s

D. 0 m/s



Part II: Give short and precise answer for the following questions (5%)

1. State the three Kepler's laws of planetary motion. (3pts)

① Kepler's 1st law "The orbit of each planet in ~~the~~ the solar system is an ellipse. The sun will be the one focus"

② Kepler's 2nd law "The radius vector connecting the center of the sun and the planet sweeps out equal areas in equal interval of time"

③ Kepler's 3rd law "The square of the orbital period of a planet is proportional to the cube of the average distance b/w the center of the planet"

$$\frac{T^2}{r^3} = \text{constant} \Rightarrow \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3$$

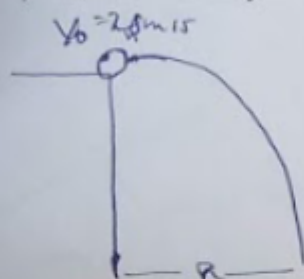
2. State Archimedes' Principle (2 pts)

~~The amount of mass submerged is equal to the mass of the displaced fluid.~~
Buoyant force of an object is equal to the weight of the displaced fluid.

Part III: Solve the following problems by showing all the necessary steps (16%)

1. A ball is tossed from an upper-story window of a building. The ball is given an initial velocity of 25 m/s at an angle of 53° above the horizontal. It strikes the ground 3 seconds later. Use: $g = 10 \text{ m/s}^2$ (4pt)

- a) How far horizontally from the base of the building does the ball strike the ground?
b) Find the height from which the ball was thrown.
c) With what velocity it strikes the ground?



Q1) find Range

from kinematic equation

$$d = vt$$

but Range is the horizontal distance

so

$$dx = v_x t \Rightarrow \text{and } v_x \text{ is } v \cos \theta$$

$$\text{and } dx = \text{Range}$$

$$R = v \cos \theta t \Rightarrow \text{and velocity is constant along horizontal}$$

$$= 25 \cos 53^\circ \times 3$$

$$R = 25 \times \cos 53^\circ \times 3$$

$$= 25 \times 0.6 \times 3 = \underline{45 \text{ m}}$$

Q2) height

from kinematic equation

$$d = v_0 t + \frac{1}{2} a t^2$$

$$dy = v_{y0} t + \frac{1}{2} a_y t^2$$

$$= v_{y0} = 0 \text{ b/c on maximum}$$

horizontal velocity along y (vertical direction) it or flat means there is no vertical motion, there is only horizontal motion

$$\Rightarrow dy = \frac{1}{2} a_y t^2 \Rightarrow \text{in vertical}$$

$$a_y = g$$

$$h = \frac{1}{2} g t^2$$

$$h = \frac{1}{2} 10 \times 9 = \underline{45 \text{ m}}$$

$$v = u + at$$

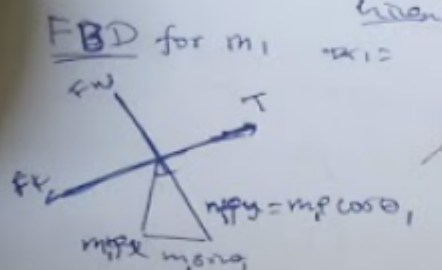
$$v = 25 + gt$$

$$= 25 + 10(3)$$

$$= \underline{55 \text{ m/s}}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

2. Two blocks of Masses $m_1 = 3.5 \text{ kg}$ and $m_2 = 8 \text{ kg}$ are connected by a massless string that passes over a frictionless pulley as shown in the figure below. The coefficient of friction between the incline and the block is 0.25. Using $g = 10 \text{ m/s}^2$, find (4pts)
- The magnitude of the acceleration of each block
 - The tension in the string.
 - Determine the direction of the acceleration.



$$\sum F_x = m_1 a = T - m_1 g \sin \theta = m_1 a$$

$$T = m_1 g \sin \theta + m_1 a \quad \text{--- (1)}$$

$$\sum F_y = 0 = F_N - m_1 g \cos \theta$$

$$F_N = m_1 g \cos \theta$$

$$T - m_1 g \sin \theta - \mu_k m_1 g \cos \theta = m_1 a \quad \text{--- (2)}$$

$$a = a = ?$$

$$T - m_1 g \sin \theta - \mu_k m_1 g \cos \theta = m_1 a$$

$$+ m_2 g \sin \theta - T - \mu_k m_2 g \cos \theta = m_2 a$$

$$m_2 g \sin \theta - m_1 g \sin \theta - \mu_k m_1 g \cos \theta - \mu_k m_2 g \cos \theta = a(m_1 + m_2)$$

$$a = \frac{m_2 g \sin \theta - m_1 g \sin \theta - \mu_k m_1 g \cos \theta - \mu_k m_2 g \cos \theta}{m_1 + m_2}$$

$$8 \times 10 \sin 60 - 3.5 \times 10 \sin 30 - 0.25(3.5 \times 10 \cos 30 + 8 \times 10 \cos 60)$$

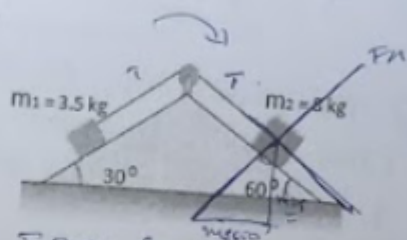
$$3.54 + 8 \text{ kN}$$

$$(8 \times 10 \times 0.86) - (3.5 \times 10 \times 0.5) - (0.25 \times (3.5 \times 10 \times 0.86) + (8 \times 10 \times 0.5))$$

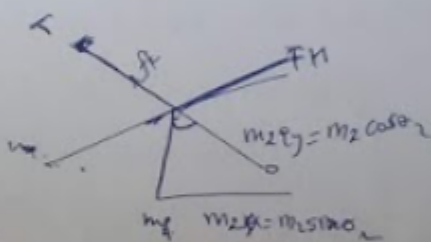
$$11.5 \text{ kN}$$

$$68.8 - 17.5 - 7.525 = 10$$

$$11.5 \text{ kN}$$



FBD for m_2



$$\sum F_x = m_2 a = m_2 g \sin \theta - T - f_k = m_2 a$$

$$= m_2 g \sin \theta - T - \mu_k F_N = m_2 a \quad \text{--- (3)}$$

$$\sum F_y = 0 = F_N - m_2 g \cos \theta = 0$$

$$F_N = m_2 g \cos \theta$$

$$m_2 g \sin \theta - T - \mu_k m_2 g \cos \theta = m_2 a$$

$$= \frac{33.775}{11.5} = 2.93 \text{ m/s}^2$$

On the back page

3. A 0.5kg sphere moving with a velocity $(2\mathbf{i} - 3\mathbf{j} + \mathbf{k})$ m/s strikes another sphere of mass 1.5kg moving with a velocity $(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$ m/s. (4pts)

- If the velocity of the 0.5kg sphere after the collision is $(-\mathbf{i} + 3\mathbf{j} - 8\mathbf{k})$ m/s, find the final velocity of the 1.5kg sphere
- Identify the kind of collision (elastic, inelastic, or perfectly inelastic).

Let $m_1 = 0.5 \text{ kg}$

$m_2 = 1.5 \text{ kg}$

(a) $m_1 \mathbf{v}_{1i} + m_2 \mathbf{v}_{2i} = m_1 \mathbf{v}_{1f} + m_2 \mathbf{v}_{2f}$

$0.5(2\mathbf{i} - 3\mathbf{j} + \mathbf{k}) + 1.5(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) = 0.5(-\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}) + 1.5\mathbf{v}_{2f}$

$\mathbf{i} - 1.5\mathbf{j} + 0.5\mathbf{k} + 1.5\mathbf{i} + 3\mathbf{j} - 4.5\mathbf{k} = -0.5\mathbf{i} + 1.5\mathbf{j} - 4\mathbf{k} + 1.5\mathbf{v}_{2f}$

$2.5\mathbf{i} + 1.5\mathbf{j} - 4\mathbf{k} = -0.5\mathbf{i} + 1.5\mathbf{j} - 4\mathbf{k} + 1.5\mathbf{v}_{2f}$

$2.5\mathbf{i} + 1.5\mathbf{j} - 4\mathbf{k} - (-0.5\mathbf{i} + 1.5\mathbf{j} - 4\mathbf{k}) = 1.5\mathbf{v}_{2f}$

$2.5\mathbf{i} + 1.5\mathbf{j} - 4\mathbf{k} + 0.5\mathbf{i} - 1.5\mathbf{j} + 4\mathbf{k} = 1.5\mathbf{v}_{2f}$

$\frac{3\mathbf{i}}{1.5} = \frac{1.5\mathbf{v}_{2f}}{1.5} \quad \mathbf{v}_{2f} = \underline{2\mathbf{i} \text{ m/s}} \quad A/4$

(b) elastic collision. (Inelastic collision)

$\Delta K_{EC} = \Delta K_{EF}$

$\frac{1}{2}m_1 \mathbf{v}_{1i}^2 + \frac{1}{2}m_2 \mathbf{v}_{2i}^2 = \frac{1}{2}m_1 \mathbf{v}_{1f}^2 + \frac{1}{2}m_2 \mathbf{v}_{2f}^2$

$0.25(4\mathbf{i}^2 + 9\mathbf{j}^2 + \mathbf{k}^2) + 2.25(\mathbf{i}^2 + 4\mathbf{j}^2 + 9\mathbf{k}^2) = 0.25(\mathbf{i}^2 + 9\mathbf{j}^2 + 64\mathbf{k}^2) + 10.75(4\mathbf{i}^2 + 2.25\mathbf{j}^2 + 0.25\mathbf{k}^2)$

$1 + 2.25 + 0.25 + 0.75 + 3.375 + 6.375 = 0.25 + 2.25 + 16.375 + 3.375$

$0.25(14) + 0.75(10) = 0.25(70) + 0.75(4)$

$3.5 + 7.5 = 17.5 + 3$

$11 \neq 20.5$

4. Water circulates throughout a house in a hot water heating system. If the water is pumped at a speed of 0.5 m/s through a 4 cm diameter pipe in the basement under a pressure of $3.03 \times 10^5 \text{ Pa}$, what will be the velocity and pressure in a 2.6 cm diameter pipe on the second floor 5 m above? (4pts)

$$V_1 = 0.5 \text{ m/s}$$

$$A_1 = 4 \text{ cm} \quad r = 2 \text{ cm} = 0.02$$

$$P_1 = 3.03 \times 10^5 \text{ Pa}$$

$$\cancel{d_2} = 2.6 \text{ cm} =$$

$$r_2 = 1.3 \times 10^{-3} \text{ m}$$

$$A_1 = \pi r_1^2$$

$$\times (2 \times 10^{-2} \text{ m})^2$$

$$A_2 = \pi r_2^2 =$$

$$A_1 V_1 = A_2 V_2$$

Equation of continuity

$$V_2 = \frac{4 \times 10^{-4} \text{ m}^2 \times 0.5 \text{ m/s}}{1.6 \times 10^{-6} \text{ m}^2}$$

$$\cancel{V_2} = \frac{4 \times 10^{-4} \times 0.5}{1.6 \times 10^{-6}} \text{ m/s}$$

$$V_2 = 7.43 \times 10^{-10}$$

$$3.03 \times 10^5 \text{ Pa} + 120 \text{ Pa} = P_2 + 120 \text{ Pa}$$

$$P_2 = 303000 \text{ Pa} - 273,293$$

$$P_2 = 29,707 \text{ Pa}$$

Bernoulli's equation

$$A_1 = \pi r_1^2 = 4 \times 10^{-4} \pi$$

$$A_2 = \pi r_2^2 = 1.69 \times 10^{-6} \pi$$

$$V_2 = \frac{1.2 \times 10^{-4} \times 0.5}{1.69 \times 10^{-6}}$$

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